20 November 1958
SUPERSEDING
NAVORD OS 2848H
19 December 1956

MILITARY SPECIFICATION

CD-10 MOD 1 MECHANISM

1. SCOPE

1.1 This specification establishes the requirements for the procurement of the clock-delay, CD-10 Mod 1 Mechanism. The mechanism, which is used to control underwater mine circuits, is a manually-wound, spring-driven clock with two normally open electrical switches which operate in sequence.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids, form a part of this specification.

SPECIFICATIONS

MILITARY

MIL-S-6872	Soldering Processes, General Specifications for.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 1350

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DRAWINGS

BUREAU OF ORDNANCE

LD 166774	CD-10 Mod 1 Mechanism.
LD 284699	Container, Inner, for Clock Delay Mechanisms.
LD 284862	Container, Outer, for Clock Delay Mechanisms.
1476194	Marking Instructions for Storage and Shipping Containers.

And all documents listed in the above LD's and drawings.

PUBLICATIONS

BUREAU OF ORDNANCE

NAVORD OP 400	General Instructions for Design, Manufacture and Inspection of Naval Ordnance Equipment.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements.

3.1.1 The clock delay, CD-10 Mod 1 Mechanism, shall be manufactured in accordance with LD 166774 and all documents listed thereon.

3.1.2 The requirements and procedures established in OP 400 shall be complied with as applicable.

3.1.3 Preproduction sample. - Unless otherwise specified in the contract or order, a preproduction sample shall be manufactured by the same procedures and processes and at the same location as proposed for full production. This sample will be tested for the purpose of determining compliance with the requirements of this specification, the drawings, and the contract. See 4.1.2.

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3.1.4 Interchangeability. - Component parts of assembled units (unless otherwise specified on the applicable drawings) fabricated in strict agreement with their applicable drawings shall be interchangeable and shall assemble with mating components of the unit without selection or fitting. If, during assembly of the fabricated parts of a unit, selection or fitting is required and changes in dimensions are necessary, no changes in the parts as delineated on applicable drawings shall be made without prior approval of the Bureau of Ordnance.

3.1.5 Conflicting requirements. - Conflicting requirements arising between this specification and any other specifications, publications, or drawings listed herein shall be referred in writing to the procuring agency or appointed agent for interpretation and clarification.

3.2 Performance requirements and product characteristics. The mechanism covered by this specification shall meet the following performance requirements and product characteristics.

3.2.1 Soldering. - Unless otherwise specified, soldering processes shall conform to the requirements specified in Military Specification MIL-S-6872. Electrical connections shall be mechanically secure before soldering wherever possible. Only rosin flux containing no acid shall be used. The use of rosin-core solder is permitted.

3.2.2 Lubrication. - Special purpose oils, which have been tested and accepted, shall be used to lubricate the Mainspring and the Gear Train. See 6.2 and 6.3.

3.2.3 Assembly nuts. - Before tightening the nuts which secure the case, the nuts for the pillar studs which protrude from the bottom plate shall be secured with a torque of 10.5 ±.5 pound-inches.

CAUTION: These nuts should not be loosened until the mechanism is ready for final assembly on a clock starter.

All other No. 8 nuts shall be secured with a torque of 10.5 ±.5 pound-inches with the exception of the cover retaining nuts which shall be secured with a torque of 4.5 ±.5 pound-inches. See 4.6.1.2.

3.2.4 Components. - The following specific requirements apply to the indicated components.

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3.2.4.1 Switches. - The switches shall comply with the following requirements, when tested as specified in 4.6.2.1.

3.2.4.1.1 Contact clearance. - While the switches are in the open position, the clearance between the contacts of the A switch shall be 0.031 to 0.050 inch and between the contacts of the C switch shall be 0.031 to 0.090 inch.

3.2.4.1.2 Contact force. - When the switches are in the open position, the inner contact arm assembly of each switch shall exert a force between 100 and 350 grams against the cam, and the outer contact arm assembly of each switch shall exert a force between 100 and 300 grams against the reinforcing angle. When the switches are in the closed position, the contact force of each switch shall be not less than 60 grams.

3.2.4.1.3 Contact resistance. - With the switches in the closed position, the resistance of each pair of contacts shall not exceed 0.1 ohm at the contact arm terminals. When measured at the terminal lugs of the cable assembly, the resistance through each switch must not exceed 0.2 ohm.

3.2.4.2 Bottom plate assembly. - There shall be less than 0.001 inch gap between pillars and the bottom plate when measured before securing the case. Measure as specified in 4.6.2.2.

3.2.5 Completed mechanism. - The following requirements apply to the completed mechanism.

3.2.5.1 Insulation resistance. - When tested as specified in 4.6.3.1, the insulation resistance shall be not less than 50 megohms:

(a) Between the contact springs of each pair of contacts when in the open position.

(b) Between each contact spring and all other contact springs.

(c) Between each contact spring and the frame of the mechanism.

3.2.5.2 Operation. - In any position, the mechanism shall operate as required by this specification. Testing shall be as specified in 4.6.3.2.

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3.2.5.3 Verge adjustment. - The verge shall be adjusted to provide the same amount of drop between the corner of the incoming pallet and the escape wheel tooth as there is between the corner of the departing pallet and the escape wheel tooth. Adjustment and testing shall be as specified in 4.6.3.3.

3.2.5.4 Locking and starting. - The mechanism shall operate when the stop lever is depressed by a force not greater than 2 1/4 pounds. The mechanism shall operate when the stop lever is 0.095 inch from the bottom plate and shall stop when the stop lever is 0.200 inch from the bottom plate. Testing and measuring shall be as specified in 4.6.3.4.

3.2.5.5 Timing. - With the clock in any position, the two switches shall have the following timing sequence as tested at the end of the clock cable.

(a) Switch A shall close in 60 minutes ± 10 percent and shall open 4 ± 2 minutes later.

(b) Switch C shall close in 170 minutes ± 10 percent.

The over-run time of the mechanism after the operation of the last switch shall be not less than 18 minutes. Switch C shall remain closed and switch A shall remain open after the mechanism has run down. Compliance with timing requirements shall be determined as specified in 4.6.3.5.

3.3 Environmental requirements.

3.3.1 Temperature. - Mechanisms covered by this specification shall be subjected to temperature conditions, for storage and operation, and tested as specified in 4.7.1. Thereafter, the mechanism shall meet all performance requirements and product characteristics of this specification.

3.3.2 Vibration. - Mechanisms covered by this specification shall be mounted and vibrated as specified in 4.7.2 and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

3.3.3 Mechanical shock. - Mechanisms covered by this specification shall be subjected to the shock test as specified in 4.7.3, and thereafter shall show no damage and shall meet all performance requirements and product characteristics of this specification.

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3.4 Workmanship and cleanliness. - The mechanisms shall be free from dust, dirt, metal filings, moisture, solder splashes, chips, and other foreign matter. Necessary precautions shall be taken during manufacture and assembly to insure these conditions.

4. QUALITY ASSURANCE PROVISIONS AND TEST REQUIREMENTS

4.1 Government inspection. - The Government inspector will make such inspections as are necessary to determine that all components and assemblies are in accordance with the requirements of the contract, pertinent specifications and drawings. Unless otherwise specified, these inspections will be conducted in accordance with instructions contained in OP 400.

4.1.1 Sampling. - Unless otherwise specified, and when applicable, the sampling plans and procedures used by the Government inspector in the determination of the acceptability of products submitted by a supplier for Government inspection will be in accordance with the provisions of MIL-STD-105.

4.1.2 Preproduction sample. - Unless otherwise specified for each contract or order, a representative sample of 24 mechanisms shall be manufactured using the methods and procedures proposed for full production. The Government inspector shall perform, witness or participate in such inspections and tests on the samples as are required for full production. Upon conditional acceptance by the inspector, the preproduction sample shall be forwarded, together with a report of inspection, to an activity designated in the contract or by the procuring agency, for the purpose of determining compliance with the requirements of the contract, specifications and drawings. Further production of the item by the contractor prior to approval of the contracting activity or the completion of the tests on preproduction samples shall be at the contractor's risk. Preproduction samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.2.1 Preproduction sample for a subsequent contract. The necessity for a preproduction sample will be determined by the procuring agency when production under a new contract by the same contractor at the same location follows the manufacture of satisfactory material as covered by this document.

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4.1.3 Lot. - As applied to the Government inspection of units of product, the term "lot" shall mean "inspection lot", i.e. a collection of units of the product used as a basis for Government inspection. Unless otherwise specified, the number of units of product in "inspection lots" shall be as determined by the Government inspector, and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.1.4 Periodic production samples. - Unless otherwise specified in the contract or order, from each five hundred (500) successively produced mechanisms, a random sample of three (3) mechanisms will be selected by the Government inspector for submission to a Bureau of Ordnance assigned activity to determine compliance with the requirements of the contract, specification and drawings. For less than five hundred (500) successively produced mechanisms the sample size shall be in accordance with standard MIL-STD-105 Appendix, Inspection Level L-4. The mechanism will be considered satisfactory if the mechanism meets all specifications and pertinent drawing requirements, and withstands the tests as specified in 4.6 and 4.7 inclusive. The failure of a mechanism to comply with any of the environmental test requirements shall result in the cessation of Government inspection until the manufacturer has corrected all deficiencies in the production process. At the discretion of the procuring activity such a failure may result in the rejection of the material represented. The manufacturer may resubmit rejected lots, subsequent to the completion of satisfactory corrective action as determined by the procuring activity.

4.2 Contractor's inspection. - Contractor's inspection shall be conducted in accordance with provisions of "Inspection by Contractor" as contained in OP 400.

4.3 Classification of tests. - The inspection and testing of mechanisms covered by this document shall be classified as follows:

(a) Acceptance tests.
(b) Preproduction tests.
(c) Periodic production tests.

4.3.1 Acceptance tests. - Acceptance tests for acceptance of the mechanism shall consist of tests to determine compliance with all requirements specified in this document except the environmental criteria.

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4.3.2 Preproduction and periodic production tests. - Preproduction and periodic production tests are those performed on samples submitted in accordance with 4.1.2 and 4.1.4 to determine that the lot and production meet all the requirements of this specification. Acceptance shall be based on no defects in a sample. Acceptable preproduction and periodic production sample units, shall not be released for service use and shall be stamped or marked as follows: "Sample - Not for Service Use".

4.4 Test equipment.

4.4.1 Test and inspection equipment and facilities. - Unless otherwise specified, the manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests under the supervision of the Government inspector. The test equipment shall be adequate in quantity, and unless definite requirements are specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

4.4.1.1 Equipment required for performing the electrical tests is as follows:

(a) Test Set Mark 95 Mod 1 or megohmmeter - 500 volts d-c range 10.0-100 megohms, accuracy within ±10 percent.

(b) One (1) four channel graphic recorder, chart speed accuracy within ±1.0 percent.

(c) One (1) ohmmeter, range 0-1 ohm, accuracy within ±5 percent.

4.5 Test conditions. - Unless otherwise specified, tests shall be performed at 80° ± 15° F and ambient relative humidity.

4.6 Acceptance tests. - Unless otherwise specified, the manufacturer shall devise procedures and methods satisfactory to the Bureau of Ordnance for conducting the acceptance tests specified herein. The right is reserved by the Bureau of Ordnance or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification, the drawings, or the contract.

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4.6.1 Visual inspection. - The mechanism and its component parts shall be inspected to verify that the materials, design, construction, physical dimensions, soldering, and workmanship are in accordance with this document and the drawings.

4.6.1.1 Lubrication. - The mechanism shall be inspected to verify that it has been lubricated as specified in 3.2.2.

4.6.1.2 Assembly nuts. - Assembly nut torque requirements specified in 3.2.3 shall be verified by the proper calibration, setting, and use of torque wrenches. Inspection shall be accomplished during the assembly process.

4.6.2 Components tests.

4.6.2.1 Switches.

NOTE: The contact arms of the two switches may be adjusted after final assembly of the mechanism, if necessary, to meet the requirements of 3.2.4.1.

4.6.2.1.1 Contact clearance. - The contact clearance of the two switches shall be checked to determine compliance with 3.2.4.1.1.

4.6.2.1.2 Contact force. - Contacts shall be checked as follows to determine compliance with 3.2.4.1.2.

(a) Open position. - The force of the inner contact arm assembly against the cam shall be measured from the end of the inner contact arm assembly. The force of the outer contact arm assembly against the reinforcing angle shall be measured from the end of the outer contact arm assembly.

(b) Closed position. - The contact force shall be measured at the outer end of the contact arm.

4.6.2.1.3 Contact resistance. - The resistance of each switch at the contacts and at the terminal lugs shall be checked with test item 4.4.1.1(c) to determine compliance with 3.2.4.1.3.

4.6.2.2 Bottom plate assembly. - After assembly of the mechanism, but before securing the case, the gap between each of the pillars and the inside face of the bottom plate shall be checked to determine compliance with 3.2.4.2.

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4.6.3 Completed mechanism tests.

CAUTION: The mechanism may be seriously damaged by over-winding. Do not use any method of winding except the key provided with the mechanism. Insert the winding key and wind only until the arrow points to the START mark on the clock housing. Do not wind beyond this point.

4.6.3.1 Insulation resistance. - A test potential of 500 ± 20 volts d-c shall be applied between circuits specified in 3.2.5.1. The insulation resistance shall be not less than 50 megohms as measured with test item 4.4.1.1 (a). To facilitate inspection, the mechanism may be tested at room ambient conditions. If the insulation resistance at ambient conditions falls below the resistance specified, the mechanism may be retested at a temperature of 77° ± 2° F with a relative humidity of 75 ± 5 percent after being subjected to these test conditions for a period of 24 hours.

4.6.3.2 Operation test. - With the mechanism operating, manually rotate it from the position with mounting studs vertically down to the position with mounting studs vertically up. Then with the mounting studs horizontal, manually rotate the mechanism through 360 degrees. Rotation in each case shall not exceed 2 rpm. Any clock which fails to operate throughout the test shall be considered defective.

4.6.3.3 Verge adjustment and test. - Adjustment of the verge, as specified in 3.2.5.3, shall be made by bending the verge. Testing of the verge shall be accomplished in the following manner. With the mainspring wound, the verge assembly shall be unlocked but shall be restrained with light finger pressure applied to the verge top weight. The verge shall be oscillated very slowly while applying light pressure toward the escape wheel along the line of centers. The drop action for the incoming and departing pallets of the verge and for each tooth of the escape wheel shall be observed visually. The locking action at both locking faces of the verge shall also be observed. Any mechanism which fails to demonstrate proper verge adjustment, as specified in 3.2.5.3, shall be considered defective.

4.6.3.4 Locking and starting tests.

4.6.3.4.1 Starting force. - A force not greater than 2 1/4 pounds shall be applied to the plunger assembly and shall unlock the verge assembly allowing the mechanism to

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start. This force shall be applied to the stop lever in a direction perpendicular to the bottom plate of the clock and along the center line of the clock.

4.6.3.4.2 Stop lever movement. - The plunger assembly shall unlock the verge assembly when the stop lever is depressed so that the distance between the bottom plate of the mechanism and the outer side of the stop lever, measured at the center of the mechanism, is not less than 0.090 inch. When the stop lever is released to a position not more than 0.200 inch from the bottom plate, measured in the same manner, the verge assembly shall be locked and the mechanism shall have stopped.

4.6.3.5 Timing test. - The clock cable shall be connected for test so that the individual leads coming from each pair of clock switch contacts are connected to separate movements of a graphic recorder, test item 4.4.1.1 (b). This will permit the recording of the operating time of each switch. The switches shall operate within the time limits specified in 3.2.5.5. During this test, each sample mechanism shall be allowed to run for 45 minutes in each of the following positions:

(a) With the mechanism mounting studs vertically up.

(b) With the mechanism mounting studs vertically down.

(c) With the verge directly above the escape wheel.

(d) With the verge directly below the escape wheel.

4.7 Environmental tests. - Mechanisms subjected to the environmental conditons given below shall be tested thereafter to determine compliance with all other requirements of this specification.

4.7.1 Temperature. - From each sample of mechanisms submitted for test, at least one mechanism shall be tested, in the following manner, to determine compliance with the requirements of 3.3.1. During this test, the rate of temperature change shall be not less than 36° F per hour. First, place the mechanism in a cold temperature chamber and decrease the temperature to -80° F and hold at this temperature for a period of 3 days. Then raise the temperature to +28° F and hold at this temperature for a period of 12 hours. With the mechanism still in the chamber at +28° F, the mechanism

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shall be tested to determine compliance with 4.6.3.5. During the timing test, pre-heat a hot temperature chamber to 120° F. Remove the mechanism from the cold temperature chamber and immediately place it in the pre-heated chamber. Now raise the temperature of the chamber from 120° F to 160° F and hold at this temperature for a period of 24 hours. Then decrease the temperature to 90° F and hold at this temperature for a period of 12 hours. With the mechanism still in the chamber at 90° F, the mechanism shall be tested to determine compliance with 4.6.3.5. When the timing test has been completed, remove the mechanism from the chamber and permit it to return to room temperature. Failure of the switches to operate within the time limits set forth in 3.2.5.5 shall be cause for rejection of the mechanism.

4.7.2 Vibration, ready-non-operating.- Mount the mechanism rigidly to the vibration table. The vertical displacement component of the vibration shall be simple harmonic motion. The samples shall be mounted on the vibration table so that equal quantities of devices experience vibration perpendicular to the plane of mounting and vibration parallel to the plane of mounting. The units shall be vibrated through a range of frequencies from 700 to 3000 cycles per minute in 100 cycles per minute increments. The duration of the vibration of each increment shall be one hour. The amplitude of the vibration shall be in accordance with the following schedule.

FREQUENCY (cycles per minute)	AMPLITUDE (inches)
700 to 1500	0.030 ± 0.002
1600 to 2000	0.020 ± 0.002
2100 to 2500	0.013 ± 0.001
2600 to 3000	0.009 ± 0.001

4.7.3 Shock, ready-non-operating. - The mechanism shall be subjected to a continuous two-phase shock applied in any direction with respect to the mechanism. The first (impact) phase shall consist of a minimum change in velocity of 5 1/2 feet per second taking place in not more than 0.4 millisecond. During the second (drag) phase, the mechanism shall be accelerated for a minimum time of 15 milliseconds in such a manner that during this time the minimum average acceleration is 90 gravity units. The peak acceleration shall be between 125 and 175 gravity units.

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4.8 Acceptance criteria. - Failure of any mechanism to meet any of the requirements of this specification shall result in rejection of the mechanism. If the number of failing mechanisms exceeds the number permitted by the associated criteria, the lot represented will be rejected.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning, drying, and preservative application. - (Not applicable).

5.1.1.2 Unit packaging. - Each clock delay mechanism shall be individually packaged in accordance with Bureau of Ordnance LD 284699.

5.1.2 Level B. - (Not applicable).

5.1.3 Level C. - (Not applicable).

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior containers. - Twelve clock delay mechanisms, individually packaged as specified for Level A protection, shall be packed in accordance with Bureau of Ordnance LD 284862.

5.2.2 Level B. - (Not applicable).

5.2.3 Level C. - (Not applicable).

5.3 Marking.

5.3.1 Special markings. - Special markings shall be in accordance with Bureau of Ordnance LD 284699.

5.3.2 Normal markings. - In addition to any special markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of Military Standard MIL-STD-129.

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5.3.2.1 For details on marking requirements, see Bureau of Ordnance Drawing number 1476194.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the title, number and date of this specification, and exceptions to this specification, applicable drawings and other documents. Procurement documents shall specify Level A packaging and packing protection for all shipments.

6.2 Oil No. 6001 as supplied by Seth Thomas Clocks of Thomaston, Connecticut, has been tested and has satisfactorily met the requirements for lubrication of the Mainspring. Any other product offered as an alternate must be tested and approved by the Bureau of Ordnance prior to use in this application.

6.3 Oil No. 5002 as supplied by Seth Thomas Clocks of Thomaston, Connecticut, has been tested and has satisfactorily met the requirements for lubrication of the Gear Train. Any other product offered as an alternate must be tested and approved by the Bureau of Ordnance prior to use in this application.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever: and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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